Durum	
Başladı	
Tamamlandı	
Süre	
Puanlar	
Not	100,00 üzerinden **57,33**

Soru 1

Doğru

2,00 üzerinden 2,00 notunu ver

Which of the following will have a compiler error?

- a. ArrayList<String> list = new ArrayList<String>();
- b. ArrayList<int> list = new ArrayList<int>(); ✓
- c. Both of these will have a compiler error
- d. Neither of these will have a compiler error

Your answer is correct.

Doğru cevap: ArrayList<int> list = new ArrayList<int>();

Soru 2

Yanlış

4,00 üzerinden 0,00 notunu ver

An abstract class (choose all that are correct)

- [x] a. Must have abstract methods ✗
- [ ] b. Cannot have abstract methods
- [ ] c. Must have non-abstract methods
- [x] d. Cannot have non-abstract methods ✗
- [x] e. Can have abstract methods ✓
- [ ] f. Can have non-abstract methods
- [ ] g. None of the other choices are correct

Your answer is incorrect.

Doğru cevaplar: Can have abstract methods, Can have non-abstract methods

Soru 3

Kısmen doğru

4,00 üzerinden 1,33 notunu ver

Which of the following data structures could be used to implement a Queue? (choose all correct responses and note, for each we are asking that if only that data structure and no other data structures be used in addition)

- [ ] a. ArrayList
- [ ] b. Array
- [x] c. LinkedList
- [ ] d. HashSet
- [ ] e. HashMap
- [ ] f. Stack
- [ ] g. TreeSet
- [ ] h. TreeMap
- [ ] i. None of these can be used since a queue is its own data structure

Your answer is partially correct.

Doğru bir şekilde seçtiniz 1.
Doğru cevaplar: ArrayList, Array, LinkedList

Soru 4

Kısmen doğru

4,00 üzerinden 1,33 notunu ver

Select all true statements about constructors

- [x] a. A default constructor is provided if none is explicitly defined
- [ ] b. A constructor is used to create an object
- [ ] c. All classes have a no-arg constructor
- [x] d. The constructor must have the same name as its class
- [x] e. A class has exactly one constructor

Your answer is partially correct.

Doğru bir şekilde seçtiniz 2.
Doğru cevaplar: A default constructor is provided if none is explicitly defined, A constructor is used to create an object, The constructor must have the same name as its class

Soru 5

Yanlış

4,00 üzerinden 0,00 notunu ver

In the shown code, there was a compiler error: Briefly explain why in the space provided:

```
class Exam {

   String examName;

   public Exam(String examName) {


      this.examName = examName;
   }
}
class FinalExam extends Exam {




}
```

Cevap:

Doğru cevap: Exam only has the constructor with a String but FinalExam tries to call a constructor with no arguments.

Soru 6

Tamamlandı

8,00 üzerinden 0,00 notunu ver

For the code in the previous question, write the corrected code by filling in the gaps with code to remove the compiler error. (Hint, you will not need to change any of the existing lines)

Yorum:

Soru **bilgi**

I gave the following code to four students for them to fill in the blanks. They could only use the String class or create their own class called Butunleme. Then when I ran each of their code I got the shown output. Based on what you know, answer the following questions:

	Code	Output

1	`public static void main(String[] args) {`
	`_____ refVar1 = _____;`
	`_____ refVar2 = _____;`
	`System.out.println(refVar1 == refVar2);`
	`System.out.println(refVar1.equals(refVar2));`
	`}`
		true
		false

2	`public static void main(String[] args) {`
	`_____ refVar1 = _____;`
	`_____ refVar2 = _____;`
	`System.out.println(refVar1 == refVar2);`
	`System.out.println(refVar1.equals(refVar2));`
	`}`
		false
		true

3	`public static void main(String[] args) {`
	`_____ refVar1 = _____;`
	`_____ refVar2 = _____;`
	`System.out.println(refVar1 == refVar2);`
	`System.out.println(refVar1.equals(refVar2));`
	`}`
		true
		true

4	`public static void main(String[] args) {`
	`_____ refVar1 = _____;`
	`_____ refVar2 = _____;`
	`System.out.println(refVar1 == refVar2);`
	`System.out.println(refVar1.equals(refVar2));`
	`}`
		false
		false

Soru **7**

Doğru

4,00 üzerinden 4,00 notunu ver

Which of the shown students must have created an extra class that does not already exist?

- [x] a. Student 1
- [ ] b. Student 2
- [ ] c. Student 3
- [ ] d. Student 4
- [ ] e. None of the above...all of them can get their outputs using existing classes

Your answer is correct.

Doğru cevap: Student 1

Soru 8

Doğru

4,00 üzerinden 4,00 notunu ver

Fill in the blanks for student 1. Only type what would go in the blanks.

Cevap:

Doğru cevap: MyClass new MyClass( ); MyClass refVar1

Yorum:

Soru 9

Doğru

4,00 üzerinden 4,00 notunu ver

Fill in the blanks for student 2. Only type what would go in the blanks.

Cevap:

Doğru cevap: String new String(" "); String new String(" ");

Yorum:

Soru 10

Doğru

4,00 üzerinden 4,00 notunu ver

Fill in the blanks for student 3. Only type what would go in the blanks.

Cevap:

Doğru cevap: String new String(" "); String refVar1;

Yorum:

Soru 11

Doğru

4,00 üzerinden 4,00 notunu ver

Fill in the blanks for student 4. Only type what would go in the blanks.

Cevap:

Doğru cevap: String new String(" "); String new String(" ");

Yorum:

Soru 12

Tamamlandı

4,00 üzerinden 4,00 notunu ver

Write a possible implementation for Butunleme for those students who used it. If multiple students used Butunleme and their implementations are different, show in the comments which student's Butunleme this is.

```
class Butunleme{
    public boolean equals(){
    return false;
   }
}
```

Yorum:

Soru **bilgi**

Due:
Tuesday, 17-June-2025 by 14:00
Deliverables:
The following Java file should be submitted to moodle by the due date and time specified above. Submissions received after the deadline will not be accepted.

- University.java

Specifications:
Overview: We want to create a class that operates as a university course registration system. Your java class will be able to register students for courses and assign grades.
Requirements: Write and modify the following classes:

1. University
    1. Attributes
        1. You will create the attributes that you think are necessary
    2. Methods
        1. Constructors that create an empty university registration
            1. The key and value datatypes must be defined (see the examples below)
            2. One constructor will take a parameter for the university name
            3. Another constructor will take no parameters and use the name "Unnamed University"
        2. getName()
            1. Returns the name of the university
        3. register(student, course)
            1. Returns none
            2. Adds the student to the given course, but does not assign a grade for this student in this course
        4. grade(student, course, grade)
            1. Returns none
            2. If the student has been registered for this course, assigns the grade for this student
            3. If the student is not registered for any courses, throws a StudentNotFoundException
            4. If the student is not registered for this course, throws a CourseNotFoundException
        5. students(course)
            1. Returns a collection of students in the course that is passed
        6. transcript(student)
            1. Returns a String of the student's courses
            2. The first line must be "Transcript for student " + {student} (note: this could be an ID or a name)
            3. Each line after the first is the format of {course} + " - " + {grade}
            4. If no grade has been assigned yet, the grade should display as "N/A" for that course
        7. toString()
            1. Returns a string representation of the University
            2. The first line must be {University name } + " information"
            3. After this, each student's transcript should be displayed
2. StudentNotFoundException is a child of RuntimeException
    1. Attributes
        1. You will create the attributes that you think are necessary
    2. Methods
        1. Constructor that takes an Object parameter for the student that was not found
        2. toString()
            1. Returns a String of the form "StudentNotFoundException - Student = " + {student}
3. CourseNotFoundException is a child of RuntimeException
    1. Attributes
        1. You will create the attributes that you think are necessary
    2. Methods
        1. Constructor that takes an Object parameter for the course that was not found
        2. toString()
            1. Returns a String of the form "CourseNotFoundException - Course = " + {course}

I have provided a main method for you to test your code. If your code does not compile with the main method, you will likely receive 0 on this portion of the exam. Do not remove any lines from the main method provided, though you may add additional lines for testing purposes. Note that the first university stores students using numerical IDs, courses as Strings and grades as integers. The second university stores student names, course numerical IDs, and grades as letters.

Using the main method provided in the file, the following output should be experienced:

Akdeniz Universitesi
StudentNotFoundException - student = 12345
CourseNotFoundException - course = CSE204
Transcript for Student 2345
CSE102T - 89
CSE204 - N/A

StudentNotFoundException - student = 23456
Akdeniz Universitesi information
Transcript for Student 1234
CSE102T - 100
CSE204 - 23

Transcript for Student 56789
CSE102T - 65

Transcript for Student 2345
CSE102T - 89
CSE204 - N/A

Transcript for Student 3645
CSE102T - 95

[1234, 56789, 2345, 3645]
[1234, 2345]
[]
Unnamed University
Unnamed University information
Transcript for Student Joseph
101 - N/A

Unnamed University information
Transcript for Student Joseph
101 - BB

University.java

```java
import java.util.*;

public class University {
   public static void main(String[] args) {
      University<Long, String, Integer> akdeniz = new University<Long, String, Integer>("Akdeniz Universitesi")

      System.out.println(akdeniz.getName());
      akdeniz.register(1234L, "CSE102T");
      akdeniz.register(56789L, "CSE102T");
      akdeniz.register(2345L, "CSE102T");
      akdeniz.register(3645L, "CSE102T");
      akdeniz.register(1234L, "CSE204");
      akdeniz.register(2345L, "CSE204");

      akdeniz.grade(1234L, "CSE102T", 100);
      akdeniz.grade(56789L, "CSE102T", 65);
      akdeniz.grade(2345L, "CSE102T", 89);
      akdeniz.grade(3645L, "CSE102T", 95);
      akdeniz.grade(1234L, "CSE204", 23);
      try {
         akdeniz.grade(12345L, "CSE204", 67);
      } catch(RuntimeException ex) {
         System.out.println(ex);
      }
      try {
         akdeniz.grade(3645L, "CSE204", 67);
      } catch(RuntimeException ex) {
         System.out.println(ex);
      }

      System.out.println(akdeniz.transcript(2345L));
      try {
         System.out.println(akdeniz.transcript(23456L));
      } catch(RuntimeException ex) {
         System.out.println(ex);
      }

      System.out.println(akdeniz);

      Collection students = akdeniz.students("CSE102T");
      System.out.println(students);
      students = akdeniz.students("CSE204");
      System.out.println(students);
      students = akdeniz.students("CSE101");
      System.out.println(students);

      University<String, Integer, String> auburn = new University<String, Integer, String>();
      System.out.println(auburn.getName());
      auburn.register("Joseph", 101);
      System.out.println(auburn);
      auburn.grade("Joseph", 101, "BB");
      System.out.println(auburn);
   }

```